Controlled information-preservation diagnostics for ultra-short DNA read representations

Canonical k-mers preserve high-resolution identity evidence, whereas canonical spaced-property encoding provides compact perturbation-stable auxiliary evidence for mNGS-like reads

Abstract

Clinical metagenomic next-generation sequencing (mNGS) often produces short or quality-trimmed reads, yet DNA representations are still commonly judged by downstream accuracy alone. This can obscure which information an encoding preserves before any classifier is trained. We present a controlled representation-diagnostics framework for ultra-short DNA reads and define canonical spaced-property encoding (CSP), a deterministic feature block that combines reverse-complement canonical spaced-seed counts with interpretable biochemical summaries. Across a close-relative WGS-slice grid spanning 69, 75, 100, 110, 125, 150 bp and a PE150 proxy, CSP was the top clean-perturbed stability representation in 42 of 42 length-by-perturbation settings. At 69 bp, CSP achieved mean paired cosine of 0.994 under 3% N masking and 0.988 under a 6-bp local mismatch, compared with 0.940 and 0.877 for canonical 5-mers. This advantage did not translate into universal species or resistance accuracy: canonical k-mers remained strong high-resolution identity baselines in close-relative and ARG/SNP probes. A dense context-visibility diagnostic further showed that apparent read-length thresholds depend on motif position, with motif-pair visibility emerging at 130, 140, 148 or 155 bp under different placements. These results support a bounded conclusion: short-read DNA pipelines should separate exact identity evidence from compact perturbation-stable auxiliary evidence, rather than ranking representations by a single accuracy number.

Introduction

Clinical mNGS has become an important route for pathogen detection because it can detect unexpected organisms without a fixed target panel (Wilson et al., 2014; Wilson et al., 2019; Chiu and Miller, 2019). The same clinical setting also creates difficult input conditions for computational analysis. Reads may be shortened by adapter and quality trimming, contain ambiguous bases, appear from either strand, or include background and low-biomass artifacts (Martin, 2011; Bolger et al., 2014; Salter et al., 2014). These constraints make the representation layer more than an engineering detail. Before a classifier, aligner or database index can succeed, the encoding has already decided which sequence properties remain available.

Most mature metagenomic classifiers are built around exact or near-exact word evidence. Kraken, Kraken 2, CLARK, Centrifuge and Kaiju show the practical power of indexed k-mer, minimizer or translated-word matching at scale (Wood and Salzberg, 2014; Wood et al., 2019; Ounit et al., 2015; Kim et al., 2016; Menzel et al., 2016). CAMI benchmarks further show that apparent performance depends on novelty, taxonomic difficulty, abundance structure and database coverage (Sczyrba et al., 2017; Meyer et al., 2022). These observations argue against using a small local study as a clinical species-identification leaderboard. They also motivate a narrower and testable question: what information does each representation preserve under the short-read perturbations that clinical pipelines actually encounter?

Deep learning has widened the representational vocabulary for DNA. Convolutional and recurrent models have been used to learn regulatory sequence specificity (Alipanahi et al., 2015; Zhou and Troyanskaya, 2015; Quang and Xie, 2016); read-level metagenomic neural classifiers include recurrent and attention-based models (Liang et al., 2020; Wichmann et al., 2023); and DNA foundation models now include k-mer token models, efficient multi-species pretraining, long-context models, reverse-complement-aware architectures and single-nucleotide generative models (Ji et al., 2021; Zhou et al., 2024; Dalla-Torre et al., 2025; Nguyen et al., 2023; Schiff et al., 2024; Fishman et al., 2025; Nguyen et al., 2024). However, a more expressive model cannot attend to information that is absent from the observed read. The short-read setting therefore requires diagnostics that separate model capacity from input information loss.

Here we propose a controlled information-preservation study rather than a final clinical classifier. Our central claim is not that CSP replaces canonical k-mers. Instead, canonical k-mers provide high-resolution identity evidence, whereas CSP provides compact, strand-friendly and perturbation-stable auxiliary evidence. We evaluate this claim using WGS-derived close-relative reads, hospital-like 69/75 bp perturbation grids, CSP component ablation, k and spaced-pattern sensitivity, lightweight readout probes, synthetic ARG/SNP boundary tasks and attention-style context-visibility diagnostics.

Related Work

Alignment-free sequence comparison treats word content as a proxy for sequence relatedness. k-mer counting, MinHash sketches and related methods provide efficient representations for genome comparison and metagenomic classification (Marcais and Kingsford, 2011; Ondov et al., 2016; Zielezinski et al., 2017). Canonical k-mers collapse a word and its reverse complement to the same index, which is useful for strand-ambiguous reads but can remove strand-specific signals. Spaced seeds, introduced for sensitive homology search and later applied to metagenomic classification, sample non-contiguous positions within a word and can improve tolerance to mismatches (Ma et al., 2002; Brinda et al., 2015).

DNA can also be represented as a numerical signal. Shannon information theory provides a language for uncertainty and information loss (Shannon, 1948), while chaos game representation, genomic signal processing and EIIP-style mappings show that nucleotide sequences can be converted into compositional or physicochemical channels (Jeffrey, 1990; Voss, 1992; Anastassiou, 2001; Cristea, 2002; Nair and Sreenadhan, 2006). dna2vec similarly bridges discrete k-mers and continuous representations (Ng, 2017). CSP follows this tradition in a deliberately modest way: it adds interpretable biochemical summaries to canonical spaced counts instead of learning a new embedding from large corpora.

Transformer models add the separate issue of token position and co-occurrence. Self-attention can connect all observed tokens (Vaswani et al., 2017), and rotary position embeddings provide a compact relative-position mechanism (Su et al., 2021). But attention cannot recover a motif that is outside the sequenced fragment. We therefore include an attention-context diagnostic that measures visibility of motif relations before attributing read-length effects to a particular neural architecture.

ARG and antimicrobial-resistance analysis imposes stricter biological requirements than coarse taxonomic assignment. Resources and tools such as CARD, AMRFinderPlus, ResFinder and MEGARes/AMR++ encode curated gene families, protein-level evidence, mutation rules and resistome workflows (Alcock et al., 2023; Feldgarden et al., 2021; Bortolaia et al., 2020; Bonin et al., 2023). DeepARG illustrates the use of learned models for ARG prediction (Arango-Argoty et al., 2018). Our experiments do not claim clinical ARG calling. They test whether a compact property-aware block can preserve perturbed ARG-like signal and where exact sequence evidence remains indispensable.

Terminology and Contribution

We use one term consistently throughout the manuscript. **Canonical k-mer** denotes reverse-complement pooled contiguous k-mer counts. **Canonical spaced seed** denotes reverse-complement pooled counts of non-contiguous tokens. **CSP** denotes canonical spaced-property encoding, implemented in code as `cspaced_property_l2`. **Hybrid** denotes a concatenation of canonical k-mer counts and CSP followed by L2 normalization. **Perturbation stability** denotes similarity between a clean read and a mutated, N-masked, trimmed or locally mismatched version of the same read. It does not mean resistance to multi-species contamination.

The contribution is therefore a representation-diagnostics framework, not a new end-to-end taxonomic classifier. The framework is designed to answer four questions: (i) which features remain stable when a short read is lightly perturbed, (ii) which priors inside CSP contribute to stability, (iii) whether identity-like readout probes favor the same representation, and (iv) when short reads structurally remove context that an attention model would need.

Method

Representation definitions

Let a DNA read be $x = x_1, \dots, x_L$, with $x_i \in \{A, C, G, T, N\}$. For a contiguous word $w = x_i, \dots, x_{i+k-1}$, ordinary k-mer counting stores $c_w(x)$. The reverse-complement canonical operation maps a word and its reverse complement to one feature index,

$$\text{canon}(w) = \min_{\text{lex}}(w, \text{rc}(w)).$$

For a spaced seed pattern $P = (p_1, \dots, p_m)$, the spaced token beginning at position i is

$$s_{i,P}(x) = x_{i+p_1} \dots x_{i+p_m}.$$

The canonical spaced count vector is $c_{\text{canon}(s_{i,P})}(x)$. CSP uses the default pattern $P = (0, 2, 4, 6)$, then concatenates a low-dimensional property vector $g(x)$. The property vector contains the mean and standard deviation of hydrogen-bond class, GC indicator, purine indicator and EIIP-like base values, plus N fraction, normalized length and Shannon entropy. The final representation is

$$\phi_{\text{CSP}}(x) = \text{L2}\left([\text{L2}(c_{\text{canon-spaced}}(x)); g(x)]\right).$$

Thus CSP is not a learned embedding. It is a deterministic auxiliary block that couples strand-canonical spaced evidence with interpretable biochemical summaries.

Canonical 5-mer and 7-mer baselines were included as high-resolution identity features. Canonical spaced seeds were included as a compact mismatch-tolerant baseline. Hybrid features were constructed as L2-normalized concatenations of canonical k-mer counts and CSP. All vocabulary-dependent features were fitted on the training or clean subset defined by each experiment to avoid using perturbed test sequences to define the vocabulary.

Datasets and perturbations

The main stage-2 WGS panel contained 21 genomes from six close or clinically relevant genera: Acinetobacter, Burkholderia, Candida, Enterobacter, Escherichia and Klebsiella. Reads were generated at 69, 75, 100, 110, 125 and 150 bp, plus a PE150 proxy represented as a 300 bp paired-end-equivalent window. Each length contained 1,680 clean reads before perturbation. Perturbations were generated with fixed random seeds and included 1% substitution, 3% N masking, 5-bp trimming, combined 1% substitution plus 3% N masking, short indels and a 6-bp local mismatch block.

The 69/75 bp analysis was treated as a hospital-like short-read setting because the user-facing project context emphasized 75 bp single-end reads and approximately 69 bp post-QC reads. This experiment measured whether a perturbed read stayed close to its clean counterpart, not whether a clinical sample with multiple organisms was classified correctly.

Metrics and readout probes

Perturbation stability was measured by paired clean-perturbed cosine similarity, paired L2 drift and nearest-clean retrieval. Compactness was measured by feature dimension and density. Readout probes used nearest centroid, logistic regression and a small scikit-learn MLP with fixed random seeds. These probes measured whether a signal could be extracted by simple models. They were not interpreted as clinical accuracy estimates.

CSP ablation separated canonical spaced counts from property additions: hydrogen-bond class, GC indicator, purine indicator, EIIP-like values, N fraction, entropy and length. Parameter sensitivity swept canonical k-mer values from k=4 to k=9 and several spaced seed patterns. The attention-context diagnostic varied read length densely from 110 to 160 bp and moved a class-defining motif pair across positions 120, 130, 138 and 145. Synthetic ARG/SNP boundary probes tested ARG-family, ARG-allele and resistance-SNP style tasks under the same perturbation logic.

Results

CSP had its clearest advantage in clean-perturbed stability

Across the WGS-slice perturbation grid, CSP was the top representation by paired clean-perturbed cosine in all 42 length-by-perturbation settings. The advantage was largest in short and locally disrupted reads. At 69 bp with a 6-bp local mismatch, CSP reached mean paired cosine of 0.988 and mean L2 drift of 0.157, whereas canonical 5-mer reached 0.877 and 0.494.

Under 3% N masking at 69 bp, CSP reached 0.994 paired cosine and 0.107 L2 drift, whereas canonical 5-mer reached 0.940 and 0.344. CSP also had far fewer features than canonical 7-mers, which used roughly 8,000 observed features in the 69/75 bp WGS grid.

Full table omitted from the PDF body for readability (20 rows, 10 columns); see the Markdown manuscript and stage-2 table files.

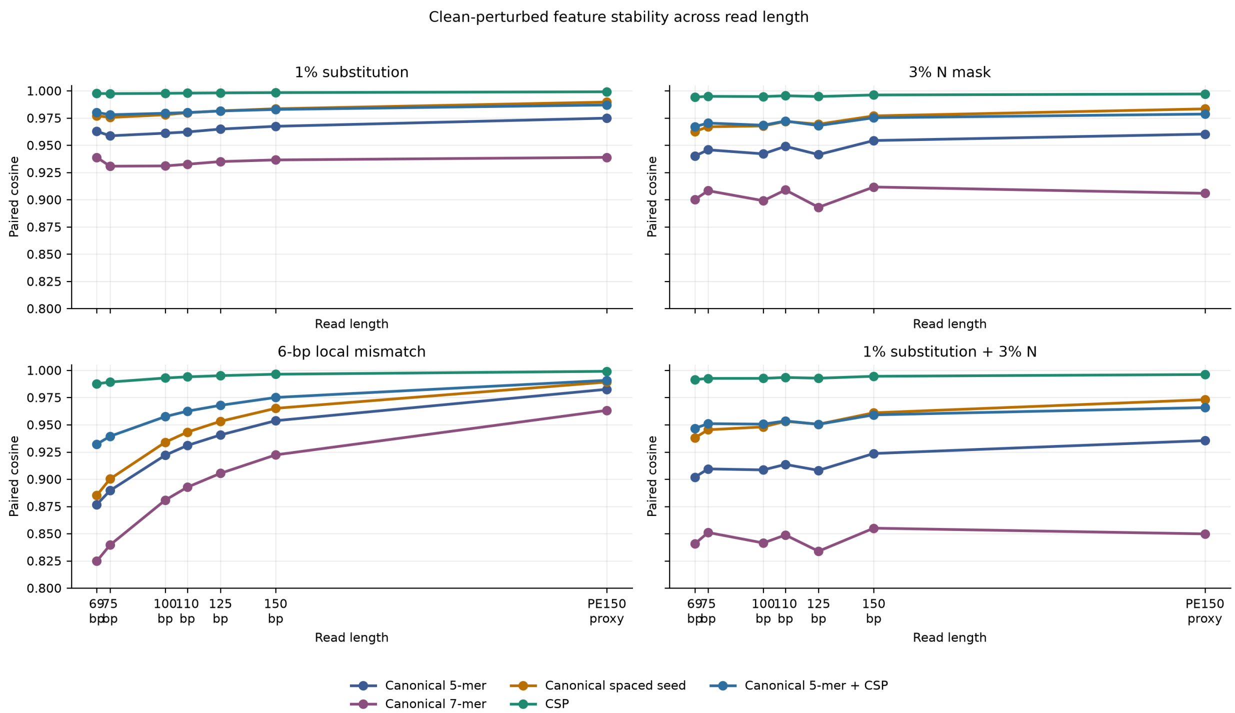

Figure 1: Figure 1. Clean-perturbed feature stability across read length.

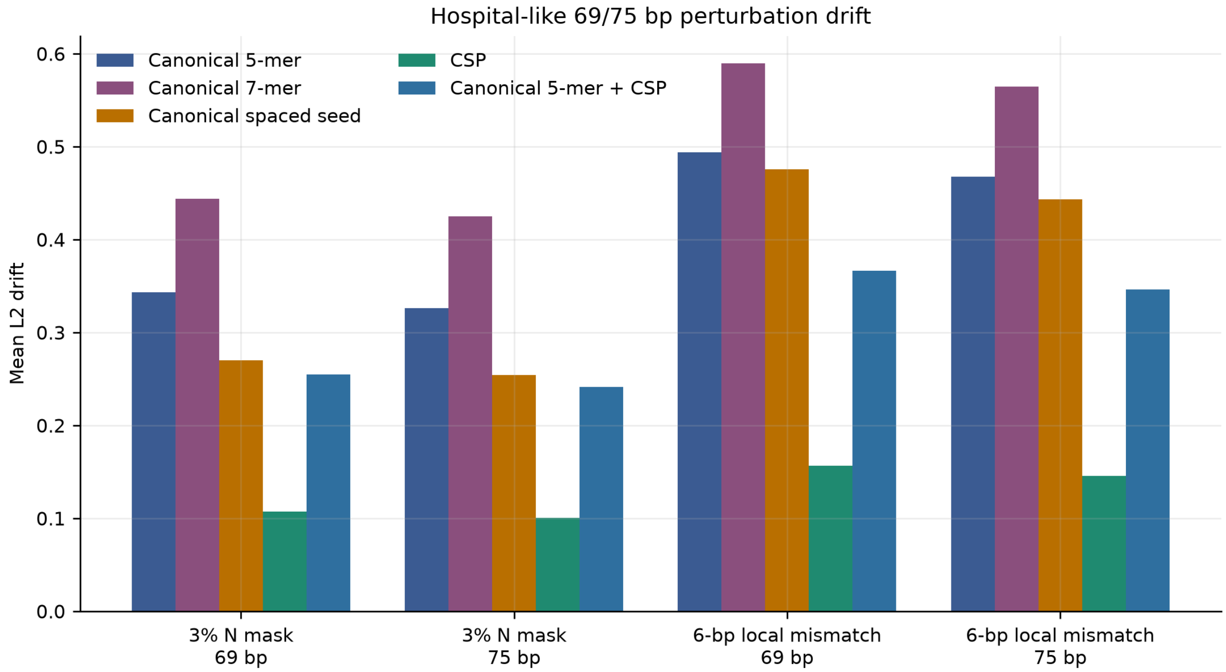

Figure 2: Figure 2. Hospital-like 69/75 bp perturbation drift.

The summary comparison confirmed that this was not a single-condition artifact. CSP exceeded canonical 5-mer, canonical 7-mer, canonical spaced seed and canonical 5-mer+CSP in paired-cosine stability in all matched comparisons.

Comparison	Mean cosine gain	Minimum gain	Maximum gain	Wins	Comparisons
CSP minus Canonical 5-mer	0.045	0.004	0.111	42	42
CSP minus Canonical 7-mer	0.08	0.008	0.163	42	42
CSP minus Canonical spaced seed	0.028	0.002	0.102	42	42
CSP minus Canonical 5-mer + CSP	0.022	0.002	0.055	42	42

Ablation showed that the property block, especially hydrogen-bond and entropy summaries, contributed to stability

CSP's stability advantage was not attributable to one isolated scalar. The full property block improved stability over canonical spaced seed counts across the tested short-read perturbations. In singleton ablations, hydrogen-bond class had the largest mean cosine gain over canonical spaced seeds, followed by entropy, purine and GC summaries. N fraction alone added little in this setup, which is expected because N masking also changes the count space.

component	Mean_cosine_gain	Mean_L2_change	Max_cosine_gain	Min_L2_change
hydrogen	0.032	-0.148	0.098	-0.3
entropy	0.016	-0.056	0.046	-0.109
purine	0.013	-0.044	0.038	-0.086
gc	0.012	-0.043	0.037	-0.086
length	0.006	-0.024	0.014	-0.055
eiip	0.001	-0.002	0.002	-0.003
n_fraction	-0	0	0	-0

Condition	Length	Cosine gain over spaced seed	L2 change over spaced seed	mean cosine	L2 drift	Features
3% N mask	69	0.032	-0.163	0.994	0.108	147
sub+N	69	0.053	-0.212	0.992	0.127	147
6-bp mismatch	69	0.101	-0.317	0.988	0.156	147
3% N mask	75	0.027	-0.152	0.995	0.1	147
sub+N	75	0.047	-0.198	0.993	0.119	147
6-bp mismatch	75	0.089	-0.297	0.989	0.146	147
3% N mask	150	0.019	-0.126	0.996	0.087	147
sub+N	150	0.033	-0.169	0.995	0.103	147
6-bp mismatch	150	0.031	-0.178	0.996	0.085	147

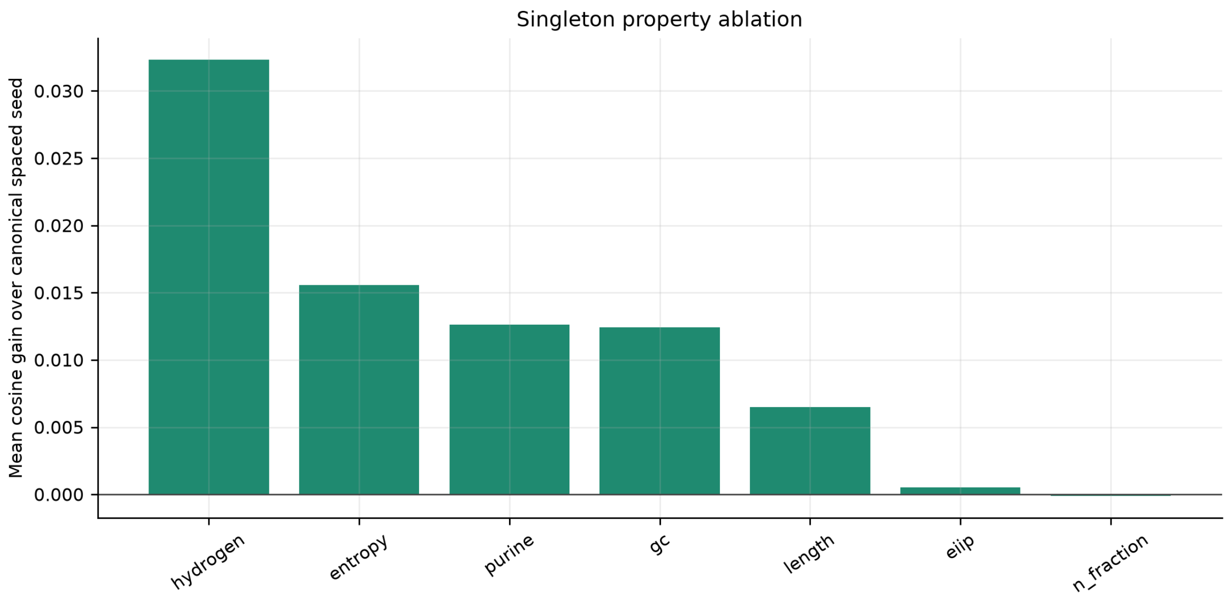

Figure 3: Figure 3. Singleton property ablation.

Lightweight readout probes separated robustness from identity resolution

Readout probes did not reproduce the stability ranking as a universal accuracy ranking. In the within-genus species probe, canonical 5-mer had the highest mean macro-F1 among the tested stage-2 representations, followed closely by canonical 7-mer and the canonical 5-mer+CSP hybrid. CSP was lower. In the target/background probe, CSP had the highest mean macro-F1 but the absolute scores remained modest. This distinction is central: CSP preserved perturbed information well, but exact k-mer evidence remained important for fine identity resolution.

Task	Representation	mean F1	accuracy	features
Target/background	CSP	0.507	0.553	146.99
Target/background	Canonical 5-mer	0.491	0.552	505.68
Target/background	5-mer+CSP	0.49	0.542	652.67
Target/background	c-spaced	0.481	0.549	135.99
Target/background	Canonical 7-mer	0.453	0.574	4395.8
Within-genus species	Canonical 5-mer	0.309	0.329	507.783
Within-genus species	Canonical 7-mer	0.307	0.331	4946.21
Within-genus species	5-mer+CSP	0.307	0.326	654.758
Within-genus species	CSP	0.301	0.319	146.975
Within-genus species	c-spaced	0.297	0.316	135.975

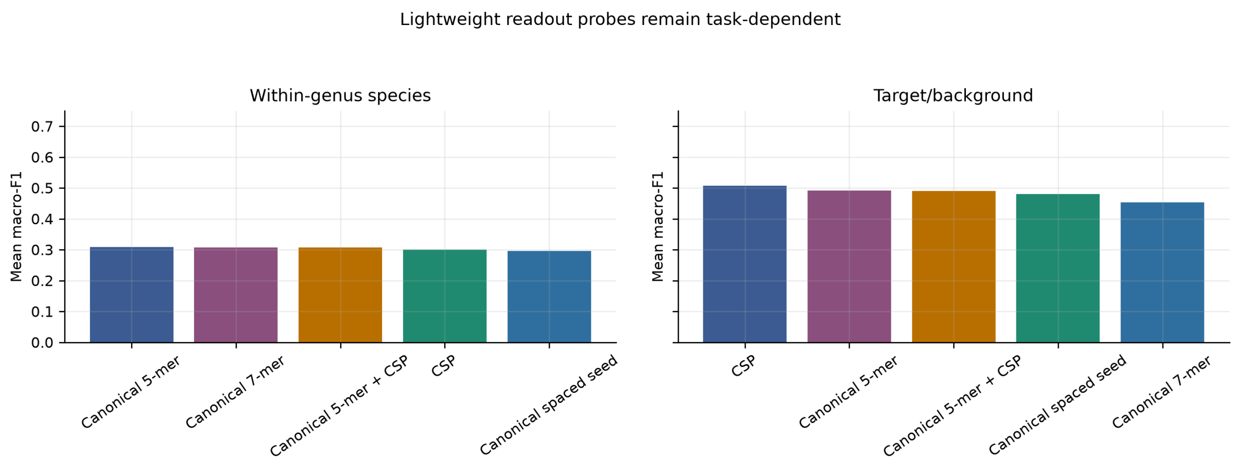

Figure 4: Figure 4. Lightweight readout probes remained task-dependent.

k and spaced-pattern sensitivity argued against a single-parameter recommendation

The parameter grid showed that the best readout configuration changed with task and read length. Target/background at 69 bp favored a canonical spaced pattern, target/background at 75 bp favored k=5, and within-genus species at 150 bp favored canonical k=6. This supports the paper's framing as a representation-diagnostics study rather than a universal prescription for one k or one seed pattern.

Probe	Length	Method	Parameter	mean F1	sd F1	Mean features
target_background	69	canonical spaced	pattern=0-2-5-7	0.555	0.085	136
target_background	75	k-mer	k=5	0.543	0.11	983.8
target_background	150	canonical k-mer	k=6	0.582	0.075	1957.2
within_genus_species	69	canonical spaced + property	pattern=0-3-5-8	0.351	0.124	147
within_genus_species	75	k-mer	k=5	0.373	0.135	979.5
within_genus_species	150	canonical k-mer	k=6	0.398	0.181	1971.33

Context loss around 125-150 bp was position-dependent, not a single read-length threshold

The attention-context diagnostic directly addressed the concern that a coarse 125 versus 150 bp comparison could misidentify a threshold. The first length with full motif-pair visibility shifted with motif placement: 130 bp when the motif pair began near position 120, 140 bp near position 130, 148 bp near position 138 and 155 bp near position 145. Thus the loss of context is not simply proportional to the number of missing bases. If a biologically or semantically relevant motif relation lies outside the read, self-attention can still connect observed tokens but cannot model the missing relation.

Motif position	first full visibility	Maximum pair visibility	Lengths tested
120	130	1	110-160 dense grid
130	140	1	110-160 dense grid
138	148	1	110-160 dense grid
145	155	1	110-160 dense grid

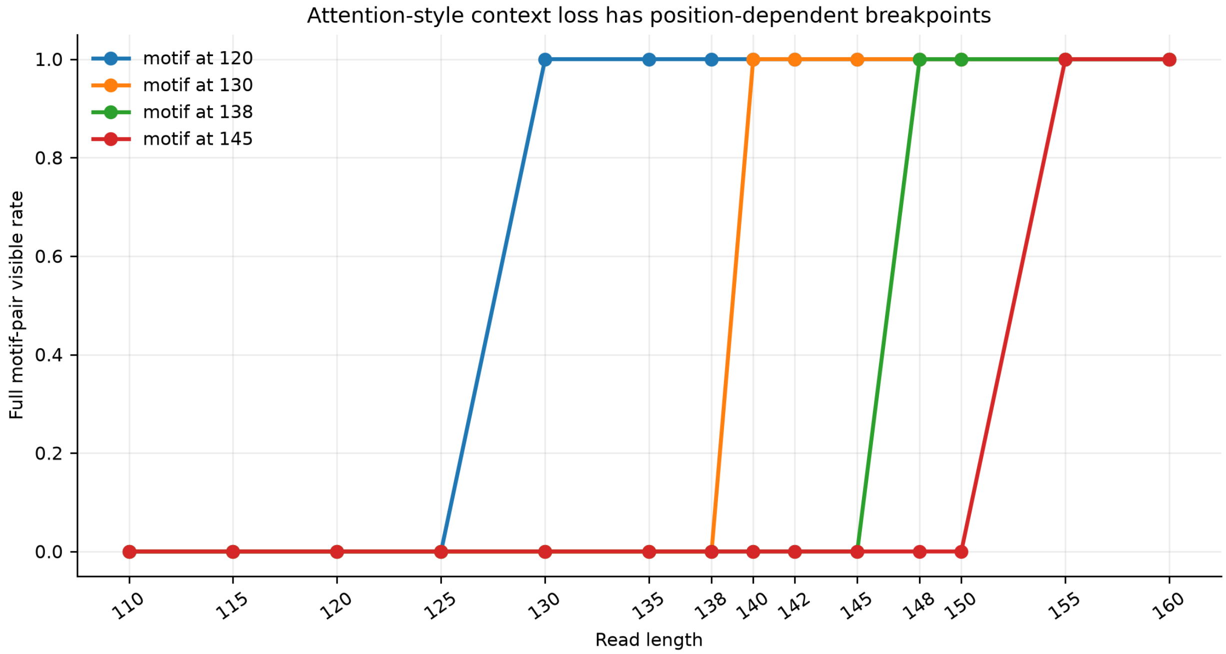

Figure 5: Figure 5. Attention-style context visibility breakpoints.

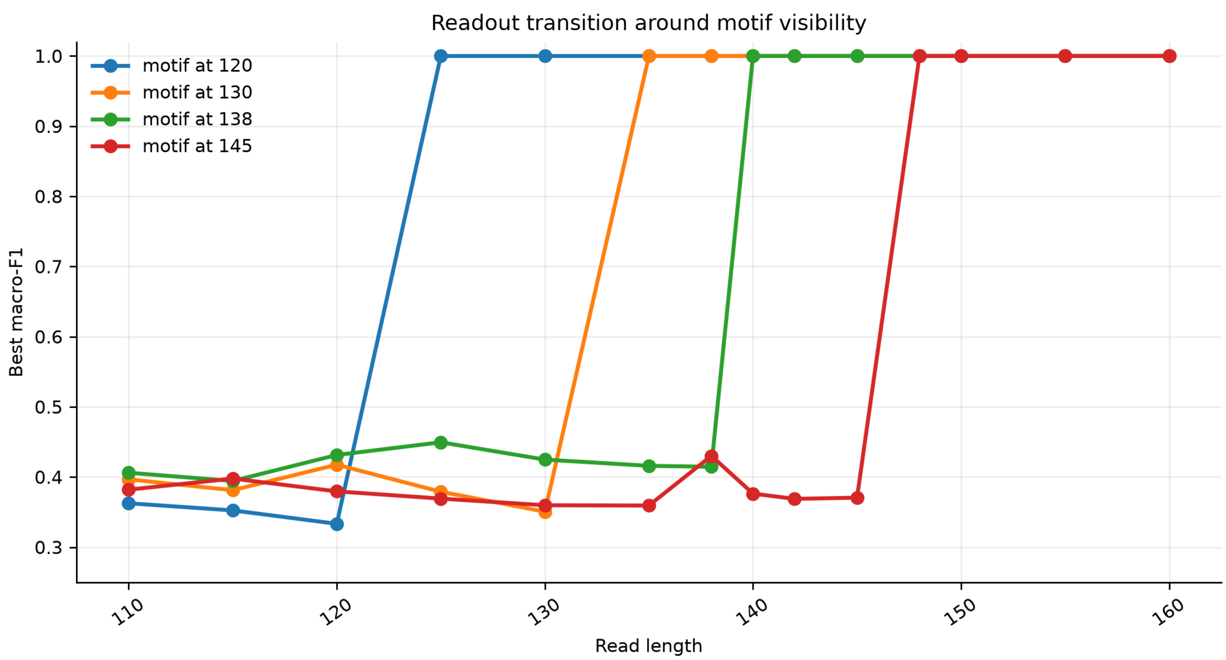

Figure 6: Figure 6. Best readout transitions around motif visibility.

ARG/SNP boundary probes bounded the role of CSP

Synthetic ARG/SNP probes showed why CSP should be treated as auxiliary evidence. CSP was again frequently the top stability representation, but the readout tasks were too easy for many representations in ARG-family and ARG-allele settings. The resistance-SNP probe showed closer differences among methods. These results do not establish clinical ARG or SNP calling. They support a weaker but useful interpretation: CSP can preserve perturbed ARG-like feature proximity, while exact k-mer, alignment or curated database evidence remains necessary for allele-level and SNP-level decisions.

Boundary task	Condition	Length	Representation	mean cosine	L2 drift	retrieval	Features
arg_allele	3% N mask	69	CSP	0.994	0.105	0.125	78
arg_allele	3% N mask	75	CSP	0.995	0.1	0.114	81
arg_allele	6-bp mismatch	69	CSP	0.99	0.139	0.119	78
arg_allele	6-bp mismatch	75	CSP	0.991	0.131	0.119	81
arg_family	3% N mask	69	CSP	0.994	0.105	0.108	101
arg_family	3% N mask	75	CSP	0.995	0.099	0.122	107
arg_family	6-bp mismatch	69	CSP	0.989	0.145	0.094	101
arg_family	6-bp mismatch	75	CSP	0.99	0.137	0.106	107
resistance_snp	3% N mask	69	CSP	0.995	0.099	0.042	56
resistance_snp	3% N mask	75	CSP	0.996	0.092	0.028	57
resistance_snp	6-bp mismatch	69	CSP	0.993	0.121	0.025	56
resistance_snp	6-bp mismatch	75	CSP	0.994	0.111	0.031	57

task	Representation	mean F1	accuracy	features
arg_allele	5-mer+CSP	1	1	407.417
arg_allele	7-mer+CSP	1	1	1160.88
arg_allele	CSP	0.999	0.999	117.875
arg_allele	Canonical 5-mer	0.999	0.999	289.542
arg_allele	Canonical 7-mer	0.999	0.999	1043
arg_allele	c-spaced	0.999	0.999	106.875
arg_family	CSP	1	1	131
arg_family	Canonical 5-mer	1	1	346.292
arg_family	5-mer+CSP	1	1	477.292
arg_family	Canonical 7-mer	1	1	1264.5
arg_family	7-mer+CSP	1	1	1395.5
arg_family	c-spaced	1	1	120
resistance_snp	5-mer+CSP	0.979	0.982	329.25
resistance_snp	CSP	0.976	0.979	97.583
resistance_snp	c-spaced	0.975	0.978	86.583
resistance_snp	Canonical 5-mer	0.975	0.978	231.667
resistance_snp	7-mer+CSP	0.973	0.977	875.667
resistance_snp	Canonical 7-mer	0.971	0.975	778.083

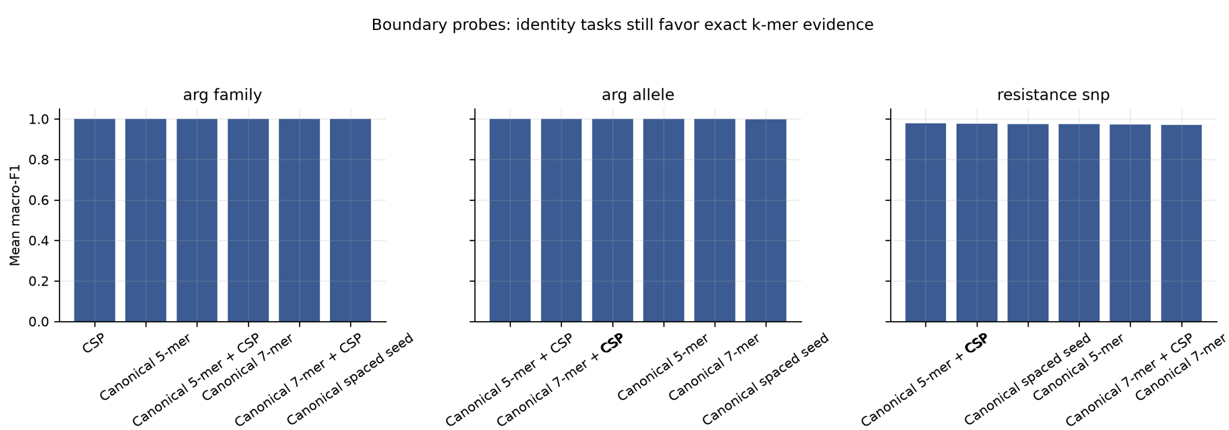

Figure 7: Figure 7. ARG/SNP boundary readout probes.

Discussion

The main result is a division of labor among representations. Canonical k-mers are still the most defensible backbone for exact identity evidence, especially when the task is close species, strain, allele or SNP resolution. CSP contributes a different property: it keeps perturbed short reads close to their clean counterparts in a compact and interpretable space. In practical terms, this means CSP is better framed as an auxiliary robustness block, a QC/audit feature or a dense side channel for downstream models, not as a replacement for canonical k-mer indices.

This distinction also resolves the apparent conflict around accuracy. Accuracy and macro-F1 are useful only as readout probes in this paper. They ask whether a simple model can extract a signal from the representation. They do not estimate clinical sensitivity, specificity or diagnostic accuracy. Overemphasizing accuracy would be misleading because the panel is intentionally controlled and small. Underemphasizing all readout probes would also be incomplete because a representation that preserves information but cannot be read by any downstream model would have limited practical value.

The most realistic future route is hybrid evidence. For mNGS species identification, canonical k-mers, alignment or database indices should provide high-resolution taxonomic evidence, whereas CSP can track whether short, N-masked or locally mismatched reads remain compositionally and biochemically near the expected clean signal. For ARG work, CSP may help characterize degraded or ambiguous reads and expose interpretable shifts, but allele calling, resistance SNP interpretation, gene context and plasmid linkage require exact sequence, protein-domain or curated database evidence.

Several conclusions remain deliberately unproven. We did not train a full Transformer or CNN on realistic clinical mNGS data. We did not benchmark Kraken2, Centrifuge, Kaiju or alignment pipelines on the same noisy FASTQ inputs. We did not use real CARD, ResFinder or AMRFinderPlus marker panels for ARG calling. These are appropriate server-stage experiments. The local experiments establish the representation-level evidence needed to justify those larger tests.

Limitations

The WGS panel contained 21 genomes from six genera and was not designed to represent microbial diversity, hospital background mixtures, abundance variation, host depletion, real quality-score distributions or wet-lab contamination. The perturbations model substitutions, N masking, trimming, short indels and local mismatches, but not full sequencer error profiles or library preparation artifacts. The readout models were intentionally small and deterministic. The ARG/SNP tasks were synthetic boundary probes. Therefore, CSP-alone species identification, ARG allele calling, resistance SNP classification, mobile-element context and plasmid linkage should not be claimed from these data.

Code and Data Availability

All code, generated lightweight reads, result tables, figures and manuscript builders are maintained in the project repository. Random seeds are fixed in the stage-2 scripts, and the earlier manuscript/results snapshot was preserved as an internal project archive before the stage-2 rerun. A public release should include the executable scripts, configuration files, generated summary tables and exact commit hash used for the submitted manuscript.

Conclusions

No single DNA representation dominated all short-read mNGS-like settings. Canonical k-mers remained the strongest general-purpose identity evidence. CSP provided a compact, strand-friendly and perturbation-stable auxiliary representation, with its strongest evidence in 69/75 bp N masking, local mismatch and combined perturbation settings. The paper's actionable message is therefore architectural rather than competitive: short-read pipelines should layer exact identity evidence with auxiliary stability evidence, and should evaluate read length through explicit context-visibility diagnostics when attention-like models are considered.

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